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Advancing first-principles dielectric property prediction of complex microwave materials: an elemental-unit decomposition approach

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Tungsten-bronze-type material $\text{Ba}_{6-3x}\text{RE}_{8+2x}\text{Ti}_{18}\text{O}_{54}$ (RE = rare earth elements) is an important microwave dielectric that has shown great promises for future miniaturization of microwave devices because of its high dielectric constant, low loss, and tunabilities, and there is still much room for improvement. With their proven predictive power, first-principles calculations may greatly help accelerate materials optimization by reducing or eliminating the expensive and time-consuming experimental trial-and-error process. However, microwave dielectrics such as the tungsten-bronze-type materials are rather complex systems with unit cells containing hundreds or thousands of atoms, making ab initio calculations prohibitively expensive. In this work, we propose an elemental-unit decomposition (EUD) technique that can drastically reduce the computational effort of predicting the properties of complex microwave dielectrics and demonstrate its accuracy and efficiency. Our approach facilitates first-principles prediction and design of complex microwave dielectric materials that would otherwise be extremely difficult.

Microwave dielectric materials play an important role in modern telecommunications^{1–7}. The ever-increasing demand for bandwidth, reliability, and miniaturization requires discovery of new high-performance microwave dielectrics, often through an expensive and time-consuming trial-and-error optimization process. Density functional theory (DFT) based modern electronic structure methods are able to accurately predict materials properties⁸, and thus may help accelerate the materials discovery process. Unfortunately, high-performance microwave dielectric materials are often rather complex. In fact, proper structural models for complex microwave dielectrics such as $\text{Ba}_{6-3x}\text{RE}_{8+2x}\text{Ti}_{18}\text{O}_{54}$ (RE = a rare earth element)^{4–6,9,10} may contain thousands of atoms in a unit cell, making first-principles calculations of the dielectric properties extremely challenging if possible at all.

$\text{Ba}_{6-3x}\text{RE}_{8+2x}\text{Ti}_{18}\text{O}_{54}$ ^{4–6,9,10} belongs to the family of pseudo tungsten-bronze-type materials which has attracted much attention due to their promising microwave dielectric properties. Despite decades of research, there is still much room for further optimization^{9–13}, and theory can potentially provide useful guidance on the optimization strategy. In addition to the vast design space by varying the composition x and the rare

earth element, there is a large amount of randomly distributed Ba and RE vacancy sites. Proper modeling of these materials requires accurate sampling of the vacancies, resulting in extremely complex structures and large supercells.

In this work, we develop an elemental-unit decomposition (EUD) method that can be used to accurately and efficiently calculate the dielectric properties of complex microwave materials, using $\text{Ba}_{6-3x}\text{RE}_{8+2x}\text{Ti}_{18}\text{O}_{54}$ as an example. In this method, the large modeling cells (which may contain thousands of atoms) are divided into smaller elemental units, with each elemental unit having the basic structure that defines the complex material but with varying composition x . These elemental units are designed to capture the essential structural and electronic features that collectively determine the properties of the material under investigation. For small to moderate size superstructures, our method is able to prediction their dielectric properties with accuracy that is consistent with straightforward DFT calculations, but at a greatly reduced computational cost. For extremely large superstructures, our method enables the predictions that are otherwise either prohibitively expensive or not possible, paving the way for design and optimization of complex microwave dielectrics from first principles.

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Results and discussion

Structural complexity of microwave dielectrics

Ba_{6-3x}RE_{8+2x}Ti₁₈O₅₄

There are two different fundamental structures of tungsten-bronze-type materials Ba_{6-3x}RE_{8+2x}Ti₁₈O₅₄⁵. One of the structures possesses an elongated cell in the *c* direction, allowing for greater freedom in tilting the octahedrons. This elongated cell, which contains 172 atoms (with *x* = 0) as depicted in Fig. 1, will serve as the fundamental structure in our discussion. However, in practical applications, the materials are much more complex. It is crucial to consider the actual superstructures with random distributions of varying amounts of Ba and RE vacancies depending on the composition *x*.

The composition *x* can be tuned from 0 to 0.75 to maintain a stable structure, as reported in previous experiments^{4,5,14}. It is evident that when *x* = 2/3, the Ba sites are fully occupied, and the superstructure contains vacancies solely on the RE sites. When *x* < 2/3, additional Ba atoms will substitute the RE sites. Moreover, if *x* > 2/3, the superstructure will have vacancies not only on the RE sites but also on the Ba sites, further complicating the structure models. Detailed descriptions of the structures with different composition *x* have been discussed elsewhere⁵. To limit our discussion to the most complex and challenging structures, we will only consider superstructures with 3/4 ≥ *x* ≥ 2/3. Table 1 shows the details of the 6 superstructures we considered in this work. For instance, when the value *x* = 2/3, the minimal superstructure comprises 3 fundamental structures, resulting in a total of 512 atoms. Within this superstructure, there are 4 vacancies located on the RE site (i.e., *V*_{RE}). But for other values of *x*, the minimal superstructures may contain thousands of atoms. These large and complex superstructures pose formidable challenges to first-principles prediction of their dielectric properties for microwave applications.

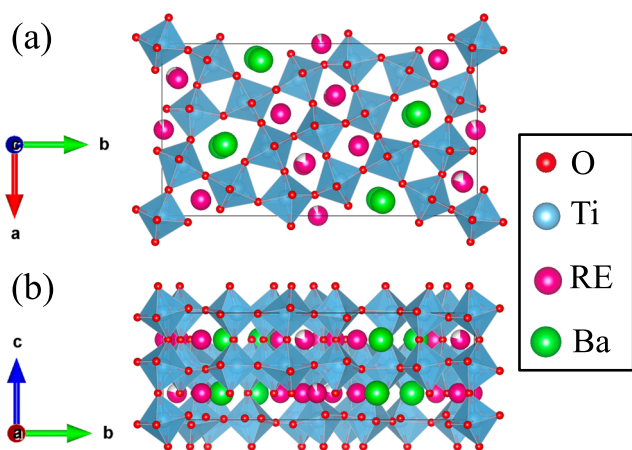


Fig. 1 | Crystal structure of tungsten-bronze-type materials. Top view (a) and side view (b) of the fundamental structure of tungsten-bronze-type materials Ba_{6-3x}RE_{8+2x}Ti₁₈O₅₄.

Elemental-unit decomposition

The most daunting challenge of first-principles calculations of the dielectric properties of microwave dielectrics comes from the extremely large size of the superstructures and the need to model the randomly distributed Ba and RE vacancies. The required computational resources for straightforward first-principles calculations of the dielectric properties of these structures will easily dwarf the most powerful computers today. Therefore, it is crucial to develop efficient and accurate approaches to address the computational difficulties.

Even though the superstructure of a given composition *x* may consist of thousands of atoms, it can be considered as an assembly of elemental units (EU), each is derived from the 172-atoms fundamental structure as discussed earlier (see Fig. 1) and has its local composition *x* as shown schematically in Fig. 2. These small EUs, depending on the type and number of the vacancies, may individually behave like an n-type or p-type semiconductor. When these defect-containing units are put together to form a whole semiconductor, self-compensation will occur. While modeling the individual EUs, we take into account the fact that the defect states (if any) are compensated without free carriers. Therefore, our EUD method and the built-in carriers compensation are not only a convenience, but they also accurately capture the self-compensation that may occur in complex microwave dielectric materials. Then the dielectric tensors ($\epsilon_{\alpha\beta}^{EU}$) of each of these small EUs can be easily calculated using the method described in the Methods section. However, even after the dielectric constants of the individual EUs have been established, there remains a necessity for an approach that enables the estimation of the effective dielectric constant for the whole superstructure.

Since the effective dielectric constant of complex structures depends on the geometric arrangement of the sub-structures. In the ideal cases where the sub-structures are arranged in parallel or in series (as shown in Supplementary Note 1 and Supplementary Fig. 1), the effective dielectric constant for electric field along the *z* direction is $\epsilon^{avg} = \frac{1}{V} \sum_i V_i \epsilon_i$ (parallel scheme) or $\frac{1}{\epsilon^{avg}} = \frac{1}{V} \sum_i \frac{V_i}{\epsilon_i}$ (vertical scheme). However, microwave dielectric are polycrystalline materials with randomly oriented grains of varying sizes. Therefore, neither of the aforementioned formulas for ideal stackings would apply. We expect that the effective dielectric constant measured experimentally of these complex structures be in between the two ideal stacking. In fact, due to relatively small variations in the dielectric constants of different elemental units, the calculated averaged dielectric constants are practically the same using these two average schemes, as will be discussed later. Thus, the dielectric tensor for the whole superstructure is evaluated by

$$\epsilon_{\alpha\beta}^{avg} = \frac{1}{V} \left(\sum_i V_i \times \epsilon_{i,\alpha\beta}^{EU} \right), \quad (1)$$

where *V* is the volume of whole superstructure, and *V_i* and $\epsilon_{i,\alpha\beta}^{EU}$ are the volume and dielectric tensor of the *i*th EU, respectively.

Take Ba_{6-3x}RE_{8+2x}Ti₁₈O₅₄ with *x* = 2/3 as an example, the smallest superstructure comprises 512 atoms, with 4 vacancies on the RE sites. This superstructure can be decomposed into three EUs, two of them have one vacancy on the RE site (denoted as EU_{V_{RE}} which contains 171 atoms) and

Table 1 | The chemical formula, minimum superstructures (*N* × fundamental structure and total number of atoms), and the vacancy sites of compositions 2/3 ≤ *x* ≤ 3/4

<i>x</i>	Full Chemical Formula	Number of Fundamental Structures	Number of Atoms	Number of Vacancies
2/3	Ba ₂₄ RE ₅₆ Ti ₁₀₈ O ₃₂₄	3	512	4 <i>V</i> _{RE}
41/60	Ba ₂₃₇ RE ₅₆₂ Ti ₁₀₈₀ O ₃₂₄₀	30	5119	3 <i>V</i> _{Ba} , 38 <i>V</i> _{RE}
42/60	Ba ₃₉ RE ₉₄ Ti ₁₈₀ O ₅₄₀	5	853	1 <i>V</i> _{Ba} , 6 <i>V</i> _{RE}
43/60	Ba ₂₃₁ RE ₅₆₆ Ti ₁₀₈₀ O ₃₂₄₀	30	5117	9 <i>V</i> _{Ba} , 34 <i>V</i> _{RE}
44/60	Ba ₁₁₄ RE ₂₈₄ Ti ₅₄₀ O ₁₆₂₀	15	2558	6 <i>V</i> _{Ba} , 16 <i>V</i> _{RE}
3/4	Ba ₁₅ RE ₃₈ Ti ₇₂ O ₂₁₆	2	341	1 <i>V</i> _{Ba} , 2 <i>V</i> _{RE}

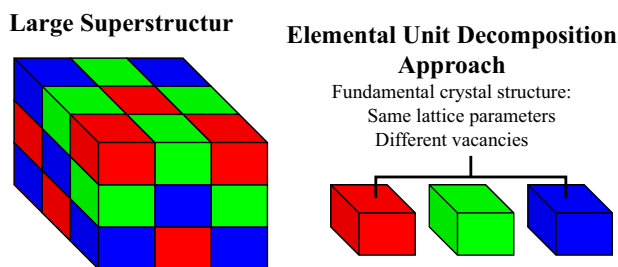


Fig. 2 | The schematic figure of the decomposition of the large crystal structure into small elemental units.

one has two vacancies ($\text{EU}_{\text{V}_{2\text{RE}}}$, which contains 170 atoms). The crystal structures of these two elemental unit structures are presented in Supplementary Fig. 2a and Supplementary Fig. 2b. The properties of these small EUs can be calculated easily using the conventional DFT approach.

The approach of decomposing complex crystal structures, particularly for disordered structures, into small constituent elements is indeed not a completely new concept. The computational demand for solving the Kohn-Sham equation in DFT calculations scales at least as N^3 , which renders the straightforward DFT calculations for large systems highly challenging. To mitigate this, the decomposition strategy has been widely utilized in the fields of condensed matter physics and materials science; these include the charge density patching method^{15,16} and the divide-and-conquer type of approach^{17–19}, among others. These previous methods aim at reconstructing the charge density of complex systems using their constituent sub-systems. These decomposition approaches significantly reduce the computational effort compared to the conventional method, thereby making the prediction of properties for complex systems more tractable.

Electronic structure: comparison between direct DFT calculation and the EUD approach

Before applying the EUD approach to calculate the dielectric properties of complex microwave materials, it is crucial to ensure that this method can reproduce the fundamental electronic structure of the material under investigation. We will use the superstructure with $x = 2/3$ to demonstrate accuracy of the EUD approach. This structure has been studied extensively and shown to have desirable dielectric properties (e.g., high dielectric constant, low temperature coefficient, and high quality factor)^{4,5,9,10}. As mentioned earlier, the superstructure with $x = 2/3$ can be decomposed into three EU (two $\text{EU}_{\text{V}_{1\text{RE}}}$ and one $\text{EU}_{\text{V}_{2\text{RE}}}$). These EUs are significantly smaller than the original superstructure. In the subsequent discussions, BNT0 (BST0) will be used to refer to the superstructure of $\text{Ba}_{6-3x}\text{Nd}_{8+2x}\text{Ti}_{18}\text{O}_{54}$ ($\text{Ba}_{6-3x}\text{Sm}_{8+2x}\text{Ti}_{18}\text{O}_{54}$) with $x = 2/3$, unless otherwise stated.

Experiment lattice constants for BNT0 and BST0 are used in our calculation^{5,14} and the atomic positions are optimized. The chosen EU crystal structures are presented in Supplementary Fig. 2. Figure 3 compares the electronic density of states (DOS) for these complex superstructures obtained using direct DFT calculations and the EUD approach. Both superstructures are semiconductors with band gaps of 2.35 eV and 2.25 eV for $\text{Ba}_{24}\text{Nd}_{56}\text{Ti}_{108}\text{O}_{324}$ and $\text{Ba}_{24}\text{Sm}_{56}\text{Ti}_{108}\text{O}_{324}$, respectively. The projected DOS shows that the top valence states are predominantly of O character (purple) whereas the low conduction states are of Ti character (as shown in Supplementary Fig. 3). Within the EUD approach, the total DOS is calculated by the summation of the DOS of all EUs. As shown in Fig. 3, the two approaches give practically identical DOS. A thorough comparison of the electronic structures among the possible configurations, each defined by a unique vacancy-vacancy (V-V) distance, is discussed in the Supplementary Fig. 4, Supplementary Fig. 5, and Supplementary Note 2. This gives us great confidence that the EUD approach can also reproduce the phonon and dielectric properties of the superstructures of complex microwave materials.

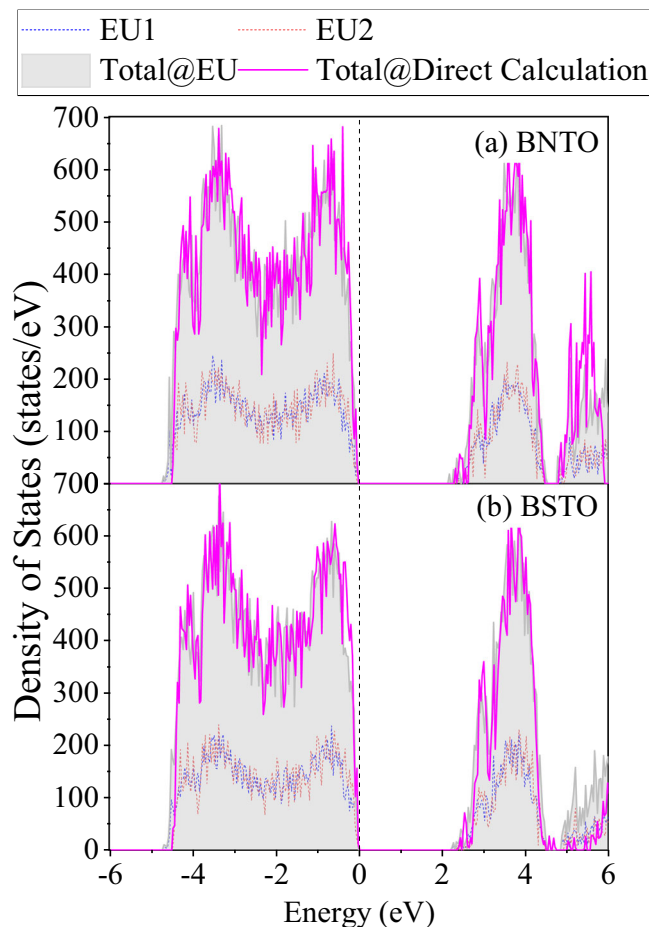


Fig. 3 | Density of states for $\text{Ba}_{24}\text{RE}_{56}\text{Ti}_{108}\text{O}_{324}$. Comparison of the density of states for $\text{Ba}_{24}\text{Nd}_{56}\text{Ti}_{108}\text{O}_{324}$ (a) $\text{Ba}_{24}\text{Sm}_{56}\text{Ti}_{108}\text{O}_{324}$ (b) obtained by direct DFT calculations and (magenta solid line) and the EUD approach (gray shadow). The density of states for the EUs are also represented by the dotted lines.

Dielectric property of BNT0 and BST0

Benefiting from the EUD method, we are now able to calculate the dielectric properties of these complex materials at finite temperatures (in this work, $T = 300$ K). We first use BNT0 and BST0 with $x = 2/3$ (which have relatively small superstructures with 512 atoms) as examples. As expected, these materials exhibit strong anisotropic properties, as shown in Table 2. The numbers in parentheses show the electronic and phonon contributions to the dielectric constants. The averaged dielectric constants are also shown in Table 2. These values can be compared with the experimental measurements⁵ since they were done on polycrystalline materials with randomly oriented grains. It is also noteworthy that, upon utilizing two distinct estimation schemes discussed above, we discerned that variations in the effective dielectric constant were only distinguishable at the second decimal place (as discussed in the Supplementary Note 1). This observation underscores the robustness and reliability of the estimation methodologies employed within our system.

In addition, it is important to emphasize that the predicted dielectric constants are about 25% lower than experimental values. This discrepancy is consistent with previous DFPT results for simple materials²⁰. Therefore, this systematic error does not come from the EUD method but rather from DFT and other details of the structural models. For example, it is well known that the calculated dielectric constants depend rather sensitively on the energy functional, e.g., generalized gradient approximation (GGA) or local density approximation functional, and the lattice constants, e.g., theoretical optimized or experimental lattice constants, used in the calculations. We have also employed the structure with lattice constants optimized using the

Table 2 | Calculated static dielectric constants of BNTO and BSTO at 300 K along the principal axes and the averaged values; experimental results⁵ are shown for comparison

Material	ϵ_a	ϵ_b	ϵ_c	ϵ_{avg}	Exp.
BNTO	67.1 (6.2, 60.9)	46.6 (6.0, 40.6)	69.2 (6.2, 63.0)	61.0	85.0
BSTO	65.1 (6.1, 59.0)	52.9 (5.9, 47.0)	62.5 (6.2, 56.3)	60.2	81.0

The electronic and phonon contributions to the dielectric constants are also shown in parentheses

Table 3 | Calculated dielectric constants along the principal axes and the averaged value for $\text{Ba}_{6-3x}\text{RE}_{8+2x}\text{Ti}_{18}\text{O}_{54}$ ($x = 43/60$ or $3/4$)

Material	Composition	ϵ_a	ϵ_b	ϵ_c	ϵ_{avg}
BNTO	43/60	66.9 (6.2, 60.7)	47.1 (5.9, 41.2)	71.4 (6.2, 65.2)	61.8
	3/4	66.9 (6.2, 60.7)	47.3 (5.9, 41.4)	72.8 (6.2, 66.6)	62.0
BSTO	43/60	60.7 (6.1, 54.5)	48.3 (5.9, 42.4)	62.2 (6.2, 56.0)	57.1
	3/4	57.9 (6.2, 51.7)	45.0 (5.9, 39.1)	62.0 (6.2, 55.8)	55.0

The electronic and phonon contributions to the dielectric constants are also shown in parentheses

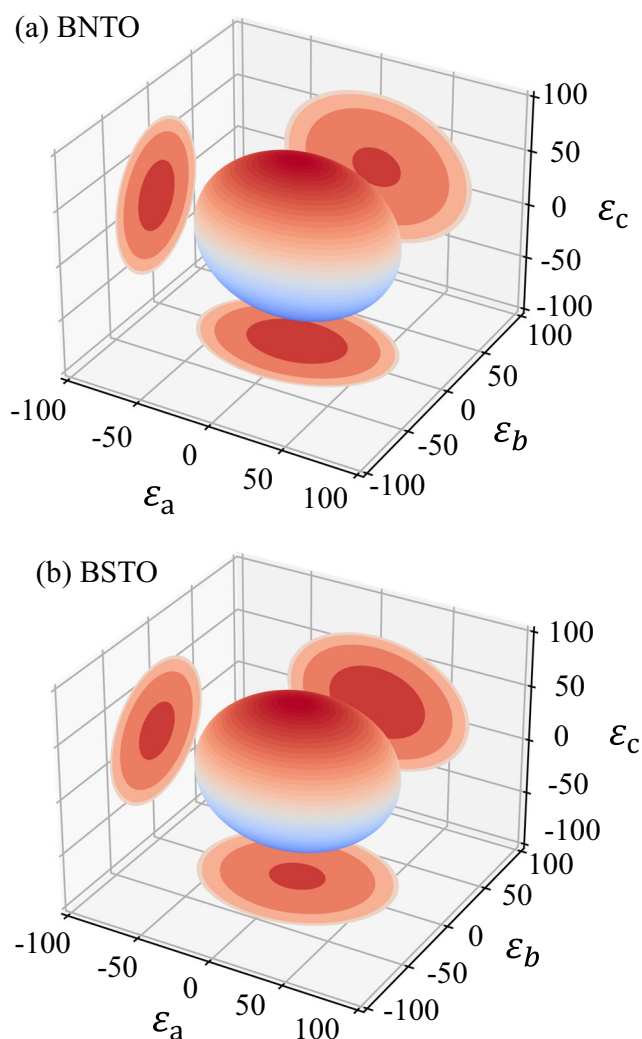
Perdew-Burke-Ernzerhof (PBE) functional to calculate the dielectric constant. Our findings reveal a notable increase in the dielectric constant results, which are thoroughly elaborated in the Supplementary Note 3 and Supplementary Table 1. Evaluating and understanding which energy functional works best for predicting the dielectric constants are also of great interest and will be the subject of further research. However, this is not the focus of current study.

Based on the structural information shown in Table 1, we proposed the following EUD: $8 \times \text{EU}_{\text{V}_{\text{IRE}}} + 13 \times \text{EU}_{\text{V}_{\text{2RE}}} + 9 \times \text{EU}_{\text{V}_{\text{1Ba}}}$ for $x = 43/60$ and $\text{EU}_{\text{V}_{\text{2RE}}} + \text{EU}_{\text{V}_{\text{1Ba}}}$ for $x = 3/4$. In doing so, the vacancy sites are evenly distributed, resulting in minimal V-V interaction. The predicted dielectric properties are shown in Table 3. The dielectric constant shows a slight reduction (increase) when x increases from $2/3$ to $3/4$ for BSTO (BNTO), which is in good agreement with the experiment results^{4,5}. The dielectric constant for systems with $x < 2/3$ can also be predicted; however, additional calculations involving different EUs should be considered.

Exploiting the anisotropy to achieve higher dielectric constants

One can also exploit the anisotropic properties to enhance the dielectric constants with texturing that orients grains along specific orientations^{9,12,21}. As we have mentioned, our EUD approach also captures the anisotropic dielectric properties. Figure 4 shows the anisotropic dielectric constant ellipsoid for the BNTO and BSTO. Notably, the principal axes a and c directions correspond to the greater dielectric constants, while the b direction corresponds to a smaller dielectric constant.

Utilizing the anisotropic property, it is possible to design the texture of ceramic to manipulate the dielectric constant of the material. The ceramics samples can be oriented in a preferred manner, either parallel to the ab plane or perpendicular to the ab plane (c direction). By controlling the weight of

**Fig. 4 | Representation of the calculated anisotropic dielectric constant results. Dielectric constant ellipsoids for $\text{Ba}_{24}\text{Nd}_{56}\text{Ti}_{108}\text{O}_{324}$ (a) and $\text{Ba}_{24}\text{Sm}_{56}\text{Ti}_{108}\text{O}_{324}$ (b).**

the aligned samples in specific directions, the variation of the dielectric constant could be evaluated by

$$\epsilon_r = (1 - x) \times \epsilon_{\text{avg}}^{\text{sc}} + x \times \epsilon_i^{\text{sc}} \quad (2)$$

which x is the weight of the aligned samples along i direction, where i is the orientation of the aligned sample and also the measured direction (which can be either the ab plane or c direction), $\epsilon_{\text{avg}}^{\text{sc}}$ is averaged dielectric constant for single-crystal (sc), and ϵ_i^{sc} is the dielectric constant along i direction. Figure 5 shows the resulting dielectric with varying weights of oriented sample. For example, in the Sm system, if 20 wt% of the ceramics samples are aligned along the c direction, the measured dielectric constant is increased by 0.76%. This result compares well with the experiment report of 0.66%^{9,10}.

It is worth noting that our EUD approach is powerful for predicting the dielectric properties of complex materials. The complexity of these materials arises solely from the concentration and distribution of defects or vacancies, which lead to an extremely large system size, making direct DFT calculations impossible. Both the electronic and phonon properties are fairly robust with respect to the concentration and distribution of the vacancies, and we do not observe any electronic or structural phase instabilities in our calculations. Within the EUD

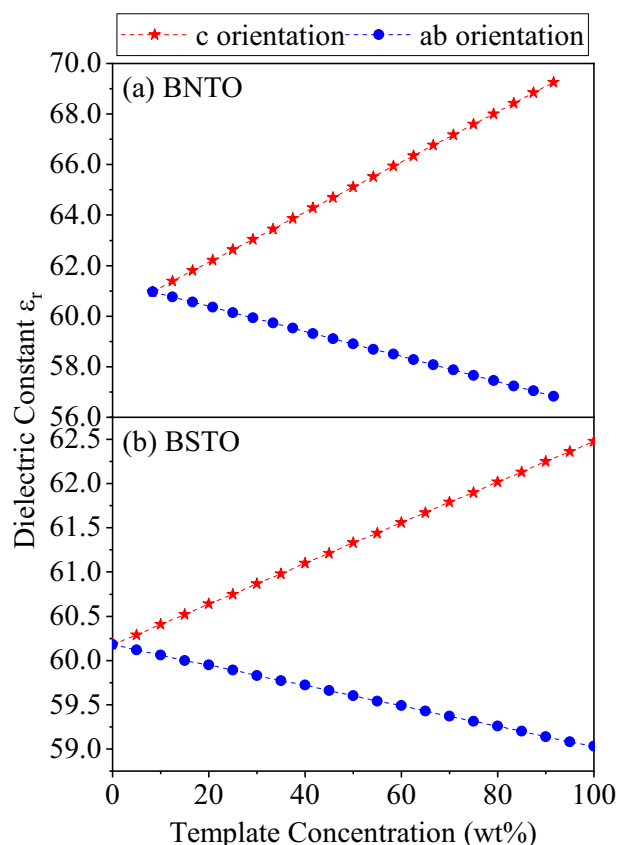


Fig. 5 | Achieving higher dielectric constant by exploiting the anisotropy. The tailing of dielectric constant for $\text{Ba}_{24}\text{Nd}_{56}\text{Ti}_{108}\text{O}_{324}$ (a) $\text{Ba}_{24}\text{Sm}_{56}\text{Ti}_{108}\text{O}_{324}$ (b) by controlling the weight of the crystalline orientations.

framework, the electronic and phonon properties of the decomposed local configurations exhibit high similarity, a result of absence of substantial local atomic displacements or phase transitions.

Nonetheless, it would be interesting to apply our EUD approach to solid solutions with mixed phases^{22–24}. In order to calculate the dielectric properties of these solid solutions, it is recommended that the phase stabilities be established first. Subsequently, the dielectric properties can be calculated for each distinct phase corresponding to specific compositions of x . If a specific composition of x for a certain phase causes the structure to become extremely complex, it is precisely under such circumstances that our EUD approach can be strategically employed.

In summary, we develop an EUD method for large scale calculations of the dielectric property of complex microwave dielectric materials using the tungsten-bronze-type material $\text{Ba}_{6-3x}\text{RE}_{8+2x}\text{Ti}_{18}\text{O}_{54}$ ($\text{RE}=\text{Nd}, \text{Sm}$) as examples. Complex microwave dielectric materials often comprise large superstructures containing thousands of atoms or more, making first-principles prediction of their dielectric properties extremely challenging. In the EUD method, the large superstructures of the complex dielectric materials are divided into small elemental units that collectively define the electronic and dielectric properties of the superstructures. The computational cost for calculating the properties of each elemental units are several orders of magnitude smaller than that is required for direct calculations of the superstructures, enabling future first-principles design and optimizations of complex microwave materials.

Methods

Dielectric constant calculation

The dielectric constant of a microwave material consists of contributions from both electron and phonon polarization, which can be calculated by

using the density functional perturbation theory (DFPT)^{8,25}:

$$\epsilon_{\alpha\beta}^{\text{total}} = \epsilon_{\alpha\beta}^{\infty} + \chi_{\alpha\beta}^{\text{ph}}, \quad (3)$$

where $\epsilon_{\alpha\beta}^{\infty}$ is the electronic contribution to the dielectric tensor, and $\chi_{\alpha\beta}^{\text{ph}}$ is the phonon (ionic) dielectric susceptibility:

$$\chi_{\alpha\beta}^{\text{ph}}(\omega) = \frac{4\pi}{\Omega_0} \sum_m \frac{\left[\sum_{\mu,k} Z_{\mu,\alpha k}^* u_m(\mu k) \right] \left[\sum_{\mu',k'} Z_{\mu',\beta k'}^* u_m(\mu' k') \right]}{\omega_m^2 - \omega^2}, \quad (4)$$

where Ω_0 is the volume of the system, $Z_{\mu,\alpha k}^*$ is the Born effective charge (tensor) of ion- μ , $u_m(\mu k)$ is the mass-normalized displacement of the m -th phonon mode on the μ -th ion in the k -th direction, and ω_m is the phonon frequency. Although DFPT has been routinely used to calculate the dielectric properties of simple materials, its straightforward application to complex microwave dielectrics faces significant challenges.

The calculation of dielectric constants using Eqs. 3, 4 necessitate knowledge of the electronic contribution to the dielectric tensor, the Born effective charge (Z^*), and the phonon properties at the zone center. The electronic contribution of the dielectric tensor can be determined by DFPT^{8,25}, or by other approaches. However, the large and complex structures often give rise to a large number of soft phonon modes (with extremely low or even imaginary frequencies) if the calculations are carried out using static structures (i.e., at zero temperature). These soft modes will naturally harden with increasing temperature. Therefore, it is necessary to perform finite-temperature phonon calculations using the temperature-dependent effective potential approach^{26–28}. To achieve this, we have employed ab initio molecular dynamics to sample the structural configurations at finite temperature, and to derive the corresponding force constants as implemented in the hiPhive package²⁹. This approach allows us to calculate phonon properties at finite temperatures. By integrating the phonon properties obtained at finite temperatures with the DFPT determined Born effective charge tensors and electronic dielectric tensors, we have effectively utilized Eqs. 3, 4 to calculate the dielectric constant for microwave materials. In addition, the extremely large superstructures of complex microwave dielectrics pose formidable challenges to straightforward DFT calculations. Our EUD method greatly alleviates the computational burden.

The DFT calculations are performed using the Vienna Ab-initio Simulation Package^{30–32}. The atomic positions are optimized until the energy and forces are converged to less than 10^{-4} eV and 0.01 eV/Å, respectively. The exchange and correlation energy are evaluated PBE functional³³ within the GGA. The plane-wave cutoff energy for wave functions is set to 500 eV.

Data availability

The datasets generated during and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Code availability

All non-commercial numerical codes to reproduce the findings of this study are available from the corresponding author upon reasonable request.

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Author contributions

Y.W. was responsible for most of the calculations and data analyses. P.Z. and W.Z. supervised the project. All authors contribute to the manuscript writing.

Competing interests

The authors declare no competing interests.

Additional information

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